



End Term (Even) Semester Regular/Back Examination May 2026

Roll no. 2412159

Name of the Program and semester: B.Tech CSE- CORE, AI-ML, AI-DS, CS | 4th Semester

Name of the Course: Finite Automata and Formal Languages

Course Code: TCS-402

Time: 3 hours

Maximum Marks: 100

Note:

- All the questions are compulsory.
- Answer any two sub questions from a, b and c in each main question.
- Total marks for each question is 20 (twenty).
- Each sub-question carries 10 marks.

Q1.

a. Construct a DFA over the alphabet $\Sigma = \{a, b\}$ that accepts all the strings from the given language L. (2X10=20 Marks) [CO2]

$L = \{a^p b^q \mid p \geq 2 \text{ and } q \text{ is a multiple of 3 like } 3, 6, 9, \dots\}$

b. Convert the given Epsilon-NFA to equivalent DFA where $\Sigma = \{0, 1, 2\}$: [CO2]

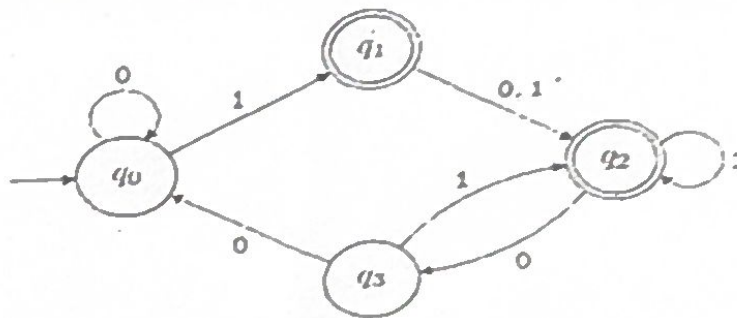
State/ symbol	ϵ	0	1	2
$\rightarrow a$	$\{b, c\}$	-	$\{b\}$	$\{c\}$
b	-	$\{a\}$	$\{c\}$	$\{a, b\}$
*c	-	-	-	-

c. Construct a DFA over the alphabet $\Sigma = \{0, 1, 2\}$ that accepts all the strings "Where every '0' is followed by a '1' and '2'". [Example of accepted strings: 1, 2, 012, 11, 22, 1012, 20121, etc.] (2X10=20 Marks) [CO2]

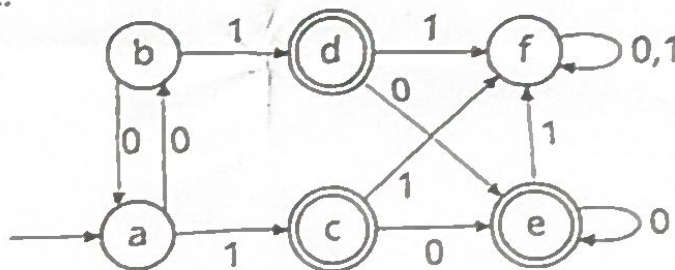
Q2.

a. State pumping lemma for regular languages. Prove that the language $L = \{0^p\}$, where p is a perfect square is not regular. [CO1]

b. Apply Arden's theorem to find the regular expression for the given finite automata: [CO2]



c. Minimize the given DFA:





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Q3.

a. Simplify the given CFG by removing useless, null and unit productions:

(2X10=20 Marks)
[CO3]

$S \rightarrow A1B \mid 1S \mid AB$
 $A \rightarrow 0A \mid \epsilon$
 $B \rightarrow 0B \mid 1C \mid \epsilon$
 $C \rightarrow 0C \mid 1C \mid \epsilon$
 $D \rightarrow 0D \mid 1D$

b. Convert the given CFG to CNF (Chomsky normal form), where $V = \{S, A, B\}$ and $T = \{a, b\}$: [CO3]

$S \rightarrow abAB$
 $A \rightarrow bAB \mid \epsilon$
 $B \rightarrow BAa \mid A \mid \epsilon$

c. Write the CFG for the given CFL:

$L = \{a^n b^m \mid n! = m, n, m > 0\}$

Further, validate the CFG by showing the production sequence for the strings $a^3 b^4$ and $a^4 b^3$ [CO3]

Q4.

a. Construct an equivalent PDA from the given CFG:

(2X10=20 Marks)
[CO4]

$S \rightarrow aAA \mid aA \mid a$
 $A \rightarrow aS \mid BS \mid b$
 $B \rightarrow aB \mid bB \mid \epsilon$

b. Design a PDA using acceptance by empty stack that accept the strings from the given language L:

$L = \{W W^R \mid W = (a+b)^*\}$

[CO4]

c. Define two-stack PDA. Design a two-stack PDA that recognize the given language L:

$L = \{a^m b^n c^m d^n \mid m \geq 2, n \geq 1\}$

[CO4]

Q5.

a. Design a Moore machine that observes a waveform having three levels a (high), b (neutral) and c (low). The machine generates a "1" whenever signal goes from b to a, generates a "0" when signal goes from b to c, otherwise generates "2". Further, convert this Moore machine to equivalent Mealy machine. (2X10=20 Marks)
[CO6]

b. Design a Turing Machine that accepts the strings from the given language:
 $L = \{x^n y^n z^n \mid n \geq 1\}$

[CO5]

c. Discuss following terms using suitable examples:

- Universal Turing machine
- Post-correspondence problem (PCP)

[CO6]